

# WILLIAM PEOCH

Lyon, France · [william.peoch@insa-lyon.fr](mailto:william.peoch@insa-lyon.fr) · +33666189377  
[williampeoch.com](http://williampeoch.com) · [GitHub](#) · [LinkedIn](#)

Research interests: medical AI, representation learning, multimodal learning, foundation models for biomedical data.

## EDUCATION

|                                                                                                                       |                                      |
|-----------------------------------------------------------------------------------------------------------------------|--------------------------------------|
| <b>INSA Lyon – National Institute of Applied Sciences</b><br>MSc in Bioinformatics and Modeling                       | Lyon, France<br>Sep 2024 – Jun 2027  |
| <b>City University of Hong Kong – Department of Mathematics</b><br>Exchange semester – Mathematics & Machine Learning | Hong Kong SAR<br>Sep 2025 – Jan 2026 |
| <b>University Institute of Technology of Reims (URCA)</b><br>Bachelor in Computer Science                             | Reims, France<br>Sep 2022 – Jun 2024 |

## RESEARCH EXPERIENCE

|                                                                                                                                    |                                     |
|------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|
| <b>INRIA (French National Institute for Research in Digital Science and Technology)</b><br><i>Machine Learning Research Intern</i> | Lyon, France<br>Jun 2025 – Aug 2025 |
|------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|

- Fine-tuned transformer-based models (BERT, LLMs) for biomedical text classification tasks.
- Built NER pipelines to extract therapeutic targets from biomedical and clinical literature.
- Collected, cleaned, and curated large-scale biomedical datasets for downstream analysis.
- Autonomous use of advanced HPC infrastructure (Grid'5000) for training ML models.

|                                                                         |                                      |
|-------------------------------------------------------------------------|--------------------------------------|
| <b>Axon' Cable (Medical Division)</b><br><i>Machine Learning Intern</i> | Reims, France<br>Apr 2024 – Jun 2024 |
|-------------------------------------------------------------------------|--------------------------------------|

- Developed deep-learning models from scratch (CNN & U-Net) to detect and segment diabetic-retinopathy lesions on retinal fundus images.
- Boosted lesion classification accuracy to 86% by applying advanced image preprocessing across diverse retinal datasets.
- Built and integrated the entire ML pipeline into a C# application now used by ophthalmologists.
- Added a RAG module to provide contextual information and improve diagnostic.

## AWARDS & COMPETITIONS

### Mistral AI × Alan Hackathon - 1st Place

Built an AI medical assistant with image diagnostics and voice/chat interface; won 1st prize out of 150+ participants and Best Pitch Award.

### Paris Bio × AI Hackathon

Fine-tuned Pixtral-12B on chest X-ray datasets to build a medical report generation system.

### LLM × Law Hackathon

Developed an IP-focused paralegal RAG system using LLM.

### Mistral Paris Hackathon

Fine-tuned Mistral-7B to generate ASCII art.

## TECHNICAL SKILLS

|                   |                                                                                     |
|-------------------|-------------------------------------------------------------------------------------|
| Machine Learning: | Deep learning, Transformers, LLMs & fine-tuning, Computer Vision, Generative models |
| Mathematics:      | Linear algebra, Probability theory, Statistics, Optimization                        |
| Bioinformatics:   | Genomics, Transcriptomics, NGS, RNA-seq, Sequence alignment                         |
| Software:         | Python, R, TypeScript, C#, Docker, Git, Linux                                       |
| Tools:            | PyTorch, TensorFlow, HuggingFace, NumPy, SciPy, scikit-learn, pandas                |

## LANGUAGES

French (native) · English (fluent, C1 – TOEIC 900/990) · Spanish (intermediate, B1) · Chinese (elementary, A2)